MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

TYPE 8128

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Forward-wave amplifier, 2.9 to 3.1 GHz operation, 60 kw peak power output, pulsed, liquid cooled, external magnetic field.

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	If(surge)	If	epy	ib	Du	tp	tk	VSWR
Unit:	V	A	A	kv	a	---	µs	sec	---
Maximum:	8.4	15	9	40	14	0.021	32	---	1.5:1
Minimum:	7.8	---	---	---	---	---	---	300	--- (Note 3)

ABSOLUTE RATINGS: See note 1

Parameter:	Waveguide pressurization	Coolant pressure, collector	Coolant temperature Inlet	Coolant temperature Bushing	Coolant flow rate, anode
Unit:	psia	psia	°C	°C	GPM
Maximum:	55	75	65	150	---
Minimum:	14 (Note 2)	---	---	---	1.5 (Note 17)

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Getter: See figure 1
Magnetic field: 750 ±50 Gauss (see notes 5 and 8)

Connectors:
Input: Type N
Output: See figure 1
Solenoid: See figure 3
Weight: 33 pounds (14.97 kg) max

TEST CONDITIONS:

Parameter:	Ef	pd	epy	Du	tp	tk	VSWR
Unit:	V	w	kv	---	µs	sec	---
Maximum:	8.0	600	37	0.021	32	---	1.1:1
Minimum:	---	500 (Note 7)	33 (Note 5)	---	29 (Note 6)	300	--- (Note 3)

Frequency	
F1	2.9 GHz
F2	3.0 GHz
F3	3.1 GHz

GENERAL:

First article sample approval - Required (See note 4)
Shelf life - See note 9

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	First article testing	4					
4278	Phase shift	13	F = F2; epy = 33 to 36 kv (see figure 2)	Δθ/Δepy	---	30	deg/kv
---	Vibration	11		---	---	---	---
1042	Shock, specified pulse	12		---	---	---	---
	Quality conformance inspection, part 1	10					
4003	Pressurizing (rf window)	-	P = 55 psia (min)	---	---	---	---
1301	Heater current	9	Ef = 8 V	If	7.7	8.7	A
4250	Power output	-	F = F1, F2, and F3	Po	1,260	2,500	W
---	Static diode impedance	5	epy as required	Z	2,550	3,150	Ohms
1143	Forced cooling	-		P	---	75	psia
	Quality conformance inspection, part 2						
1261	Electrode voltage (getter)	-	I (getter) = 15 A	E (getter)	1.0	4.0	V
4266	Direct-interelectrode capacitance	-	Anode - cathode	C	---	30	pF
1155	Coolant - pressure drop versus coolant flow (collector)	-	Flow = 1.5 GPM	ΔP	10	30	psi
4153	Liquid cooling jacket leakage	-	P = 30 psi	---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	15,16	Group D	t	1,000	---	hrs
---	Life-test end points:						
4250	Power output	-	F = F1, F2, and F3	Po	1,160	2,500	W
---	Static diode impedance	5	epy as required	Z	2,550	3,150	Ohms

NOTES: 1. In addition to MIL-E-1 symbols, the following shall apply:

P	- - - - - - - - - -	Pressure
GPM	- - - - - - - - - -	Gallons per minute
ppm	- - - - - - - - - -	Parts per million
Δθ	- - - - - - - - - -	*Change in phase*

2. The gas used in pressurization shall provide insulating properties which are the equivalent of oil-filtered and particle-filtered air dried to a dew point below -40°C. The oil trap shall be activated alumina, the filter shall be 20 to 30 micron size, and the dessicant shall be silica gel, or equivalent.

3. The VSWR presented to the TUT by the driver or load shall not exceed 1.5:1 over the frequency range of 2.7 to 2.9 GHz; 1.3:1 from 2.9 to 3.1 GHz; 1.5:1 from 3.1 to 3.4 GHz; and 2.0:1 from 3.4 to 3.5 GHz.

4. First article inspection shall consist of performing all tests listed on this tube specification sheet. The sample size and allowable defects shall be in accordance with MIL-E-1. Qualification, samples, and acceptance criteria paragraph. Three copies of the test report, prepared in accordance with MIL-STD-831, shall be forwarded to the purchasing activity for evaluation by the preparing activity.

5. The recommended value for operation shall be stamped on the identification plate of the tube.

6. The pulse characteristics shall be as follows:

trv:	0.1 (min)	1.5 (max) μs
tfv:	0.1 (min)	5.0 (max) μs

No spike or ripple shall exceed ± 3 percent of the smooth peak value of voltage or current. Inverse voltage shall not exceed 30 percent of the forward voltage.

7. The level shall remain constant at all frequencies.

8. The transverse or radial magnetic field shall not exceed 1 percent of the axial magnetic field.

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NOTES: -Continued

9. The tubes tested and procured in accordance with this tube specification sheet, shall be capable of shelf-life for a period of 12 months after date of shipment from the manufacturer's plant. Tubes placed in operation during this shelf-life period, shall meet all requirements of this tube specification sheet. Tubes found to be defective, within the shelf-life period, shall be replaced by the manufacturer at no cost to the Government, or its contractors.

10. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

11. The tube shall be mounted with the cathode axis vertical and vibrated in a nonoperation condition as follows:

Exploratory:	With a peak-to-peak amplitude and frequency range of 5 to 15 Hz at 0.075 ±0.012 inches, 16 to 25 Hz at 0.050 ±0.008 inches, 26 to 33 Hz at 0.025 ±0.004 inches.
Variable:	As specified in the exploratory test except that the dwell time at each frequency with observed resonances shall be approximately 5 minutes.
Endurance:	As specified in the variable test except that the dwell time shall be at least 2 hours duration.

The tube shall pass quality conformance inspection, part 1 upon completion of this test.

12. The TUT shall be shocked three times in a direction along the cathode axis, the shock pulse shall be a half sine wave with a duration of 11 ms and an intensity of 50 G. It shall also be mounted and shocked in the same manner for three times in the other mutually perpendicular axes, the shock pulse having a duration of 11 ms and an intensity of 25 G. The tube shall pass quality conformance inspection, part 1 upon completion of this test.

13. The $\Delta\theta/\Delta epy$ value is computed from the difference between the phase shifts through the TUT at the two anode voltage extremes divided by the total voltage change.

14. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

15. Power input to the TUT during life test shall be cycled according to the following schedule:

High voltage on:	7.5 hours
High voltage off:	0.5 hours

The rf input frequency shall be varied from F1 to F2 to F3 and repeated in intervals of 100 hours.

16. Life test required for first article testing only.

17. Water shall be used with (1) specific resistivity maintained greater than 0.5 Meg-cm, (2) oxygen content maintained less than 0.5 ppm, and (3) a 3 to 4 micron filter.

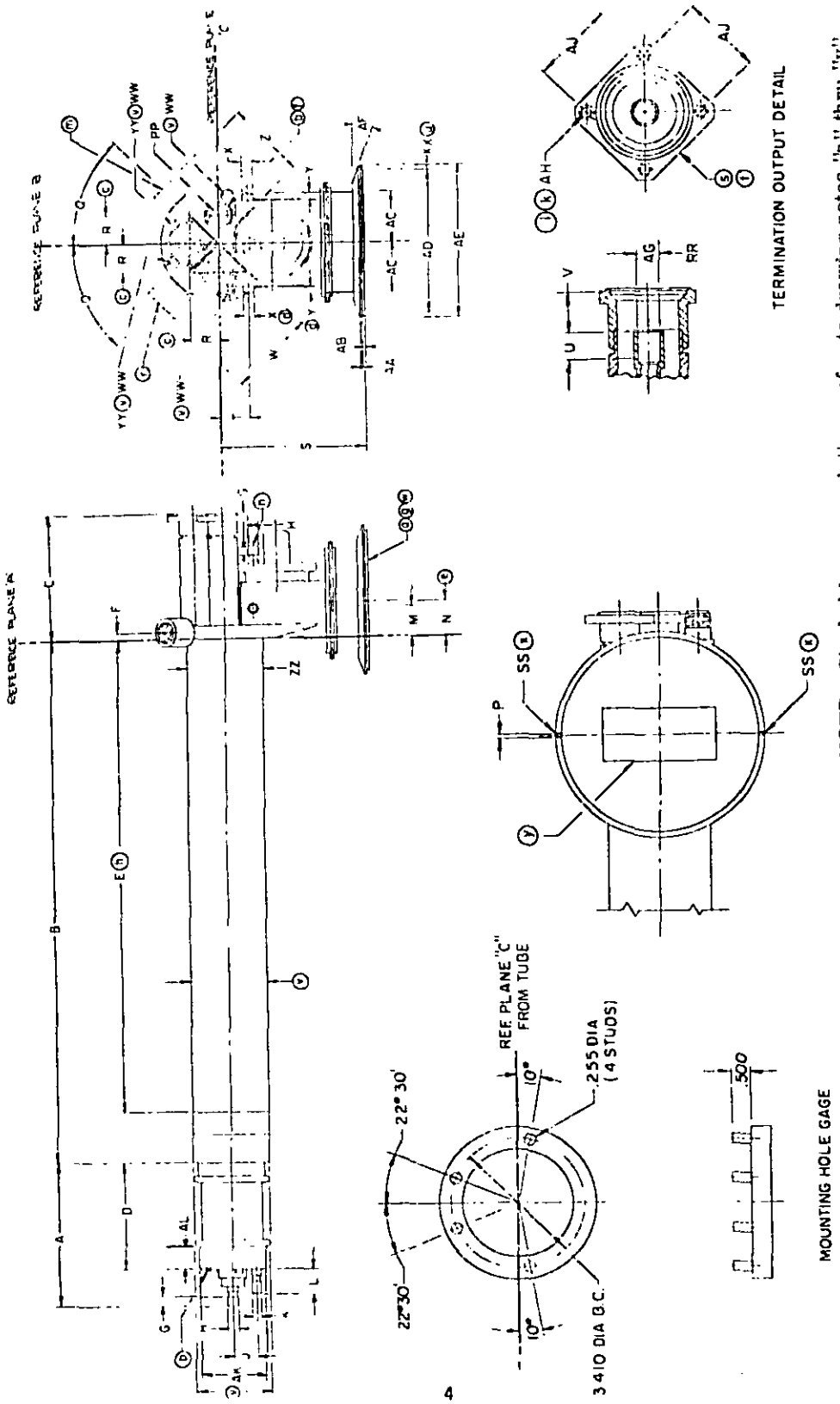
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

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NOTE: Circled lowercase letters refer to drawing notes "a" thru "y".

FIGURE 1. Outline drawing for tube type 8128.

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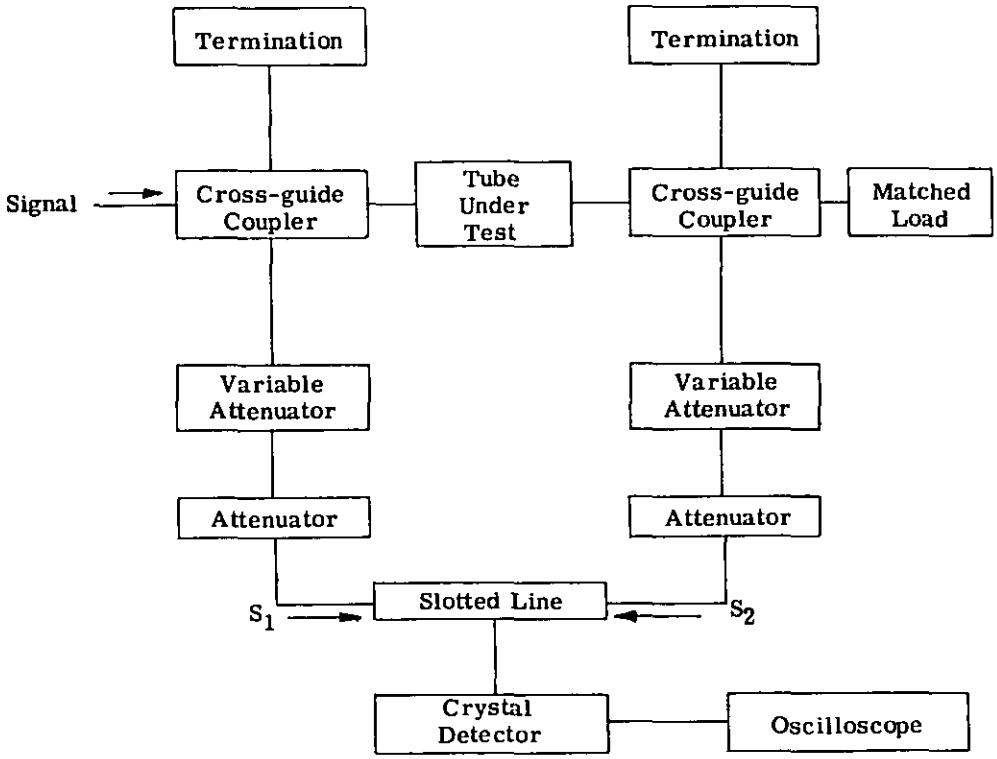
Dimensions in inches with metric equivalents (mm) in parentheses		
Ltr	Minimum	Maximum
Quality conformance inspection, part 1 (see note 10)		
A	4.867 (123.62)	5.241 (133.12)
B	18.615 (472.82)	18.950 (481.33)
C		4.455 (113.16)
N	1.200 (30.48)	1.240 (31.50)
P	.0595 (1.51)	.0655 (1.66)
S	4.960 (125.98)	5.040 (128.02)
V	.500 (12.70)	.520 (13.21)
Y	.187 (4.75)	
AG	.308 (7.82)	.316 (8.03)
Quality conformance inspection, part 2		
D	3.632 (92.25)	3.882 (98.60)
G	.312 (7.92)	
H	.367 (9.32)	.383 (9.73)
K	.306 (7.77)	.312 (7.92)
L	.855 (21.72)	.895 (22.73)
R	.950 (24.13)	1.050 (26.67)
U	.375 (9.52)	
W		3.375 (85.73)
X	.370 (9.40)	.380 (9.65)
Z		3.125 (79.38)
AB	.084 (2.13)	.104 (2.64)
AD	5.031 (127.79)	5.051 (128.30)
AE	5.285 (134.24)	5.315 (135.00)
AJ	1.006 (25.55)	1.050 (26.67)
AK	2.342 (59.49)	2.372 (60.25)
AL	.780 (19.81)	
Nominal dimensions		
E	17.125 (435.00)	
F	.235 (5.97)	
J	.850 (21.60)	
M	1.050 (26.70)	
Q	45 degrees	
T	1.000 (25.40)	
AA	.063 (1.60)	
AC	1.750 (44.50)	
AF	20 degrees	
AH	.190-32	

NOTES:
Reference plane "A" is defined as a plane passing along the face of the mounting surface as shown.
Reference plane "B" is defined as a plane perpendicular to plane "A" passing thru the axis of dia "ZZ" at plane "A" and midway between the axes of holes "YY" at plane "A".
Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing thru the axis of dia "ZZ" at reference plane "A".

a. This flange shall mate with a modified UG-54/U.
b. This part shall mate with a type "N" plug.
c. These dimensions apply to the axes of dia "RR" and "PP".
d. Terminal for ground adapter.
e. This dimension refers to the axis of dia "XX".
f. rf input (type "N").
g. rf output.
h. This section of the tube may be composed of numerous laminations, each containing a "V" notch.
k. Pitch dia shall accept a class 2 "GO" gage only.
l. Minor dia shall be not greater than .166 (4.22 mm).
m. This part shall be equivalent to AND10050-6.
n. Getter bushing.
p. Cathode temperature measurement point.
r. This part shall be equivalent to AND10050-8.
s. For termination.
t. This part shall mate with a UG-45/U coupling flange.
u. The axis of this dia shall lie on reference plane "B" within .020 (.51 mm).
v. These diameters and mounting holes "WW" shall simultaneously accept a gage 22.500 (571.50 mm) long with a 2.652 (67.36 mm) I.D. with pins as shown on "MOUNTING HOLE GAGE".
w. This surface shall be perpendicular to reference plane "A" within .020 (.51 mm).
x. A centerline created by these pins shall be parallel to reference plane "A" within .040 (1.02 mm) and pass thru axis of dia "XX" within .004 (.102 mm).
y. This surface shall be parallel to a centerline passing thru pins "SS" within .025 (.64 mm).

FIGURE 1. Outline drawing for tube type 8128 - Continued.

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Equipment required:

2 - Cross-guide couplers (30-40 dB directivity)
2 - Waveguide-to-coaxial transitions
2 - Terminations
2 - Coaxial attenuators
2 - Variable attenuators
1 - Slotted line
1 - Crystal detector
1 - Oscilloscope
1 - Matched load

For the measurement of phase sensitivity to voltage change, a setup as shown above shall be used. Both arms 1 and 2 shall be well-matched to minimize reflections. At a fixed frequency and a fixed anode voltage level, the variable attenuators shall be adjusted to equalize the magnitude of signals S_1 and S_2. (Oscilloscope comparison of the signal of S_1 with S_2 removed, and vice versa, will aid in this step.) The trace shown on the oscilloscope will be a pulse whose magnitude varies with the position of the probe carriage. The position of the minimum of the standing wave will be observed by a dip in the oscilloscope signal.

After the minimum position of the standing wave has been located, the operating point of the tube is changed by reducing the magnitude of the pulse voltage. The new minimum position shall then be located. The phase shift shall be determined from the difference of the standing wave minimum positions caused by the two voltage levels.

The pulse voltage shall be viewed on an oscilloscope to determine the two voltage levels of the measurement.

The phase sensitivity is determined by dividing the voltage change into the measured phase shift.

FIGURE 2. Phase shift variation.

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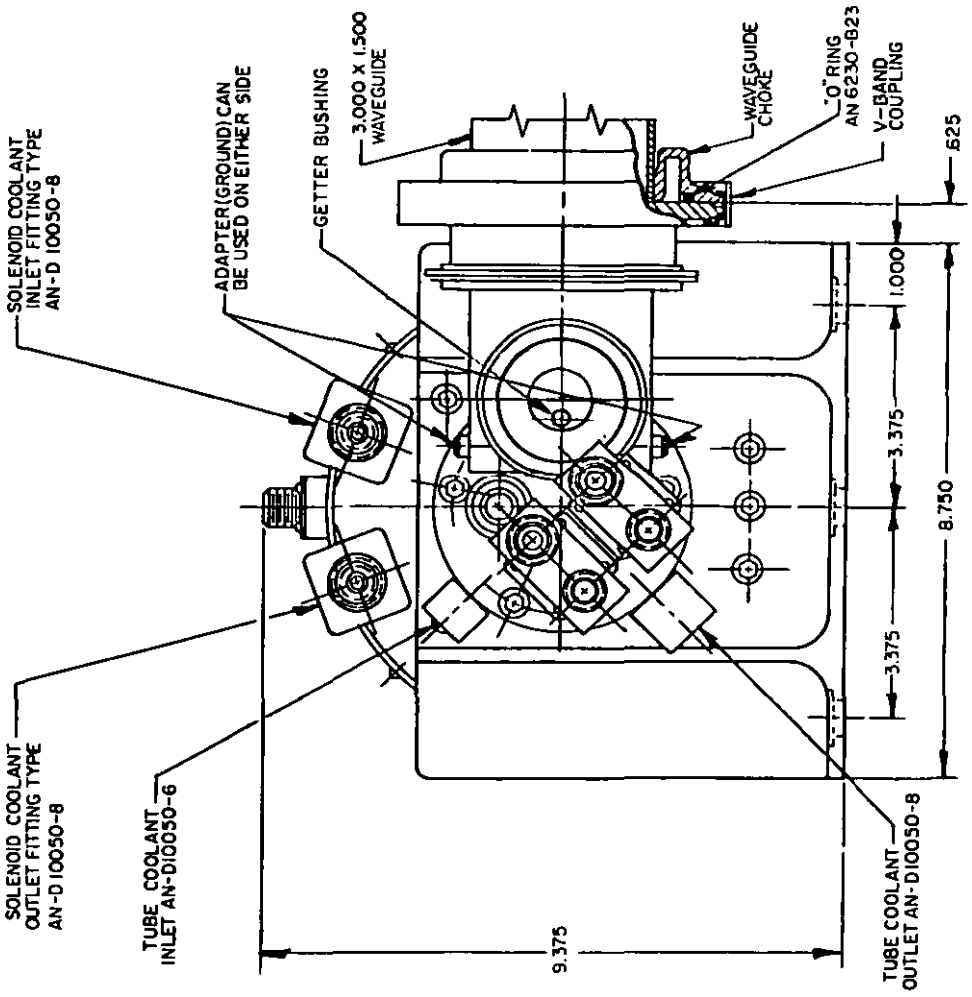


FIGURE 3. Installation drawing (for guidance only).

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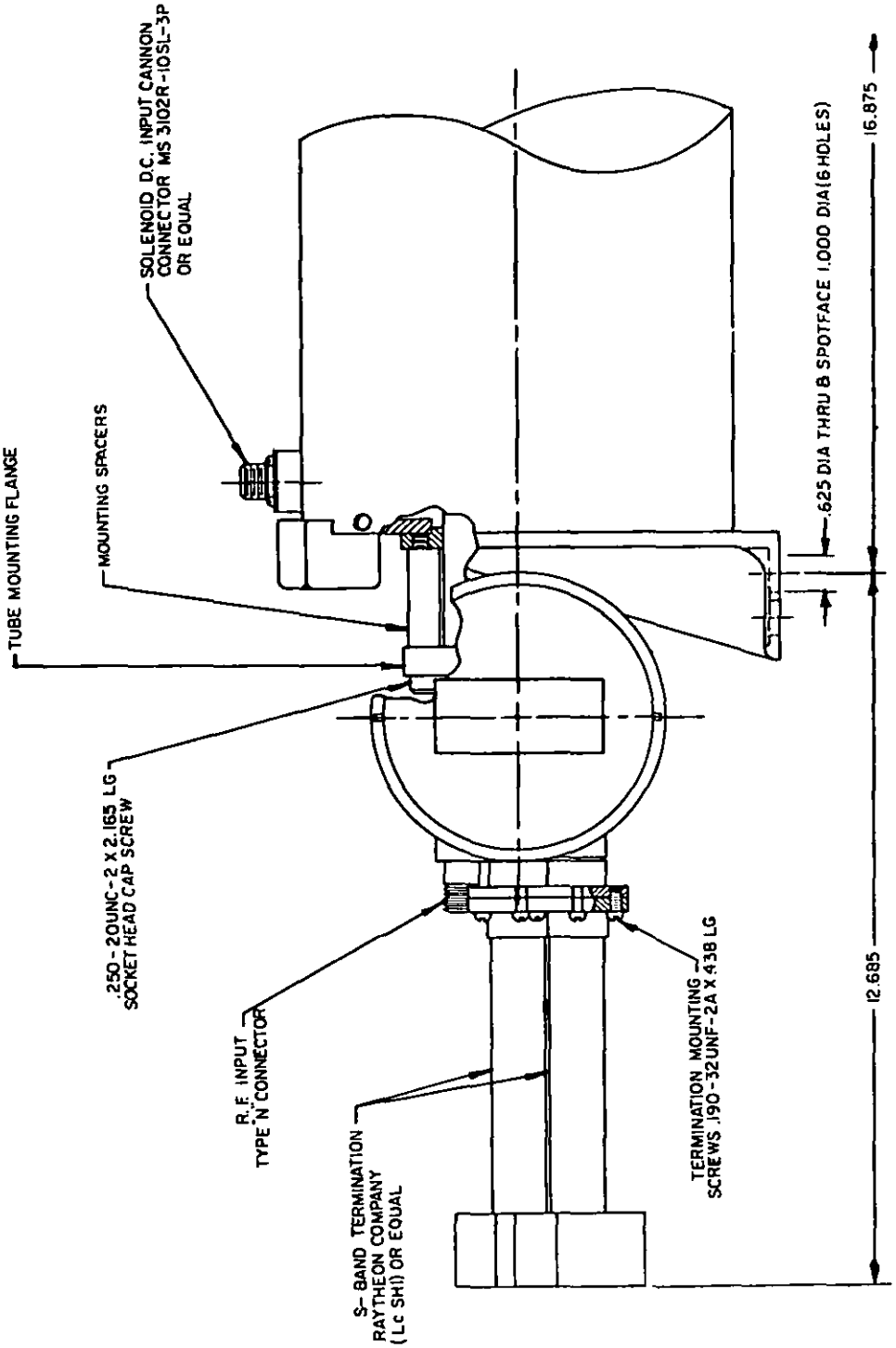


FIGURE 3. Installation drawing (for guidance only) - Continued.

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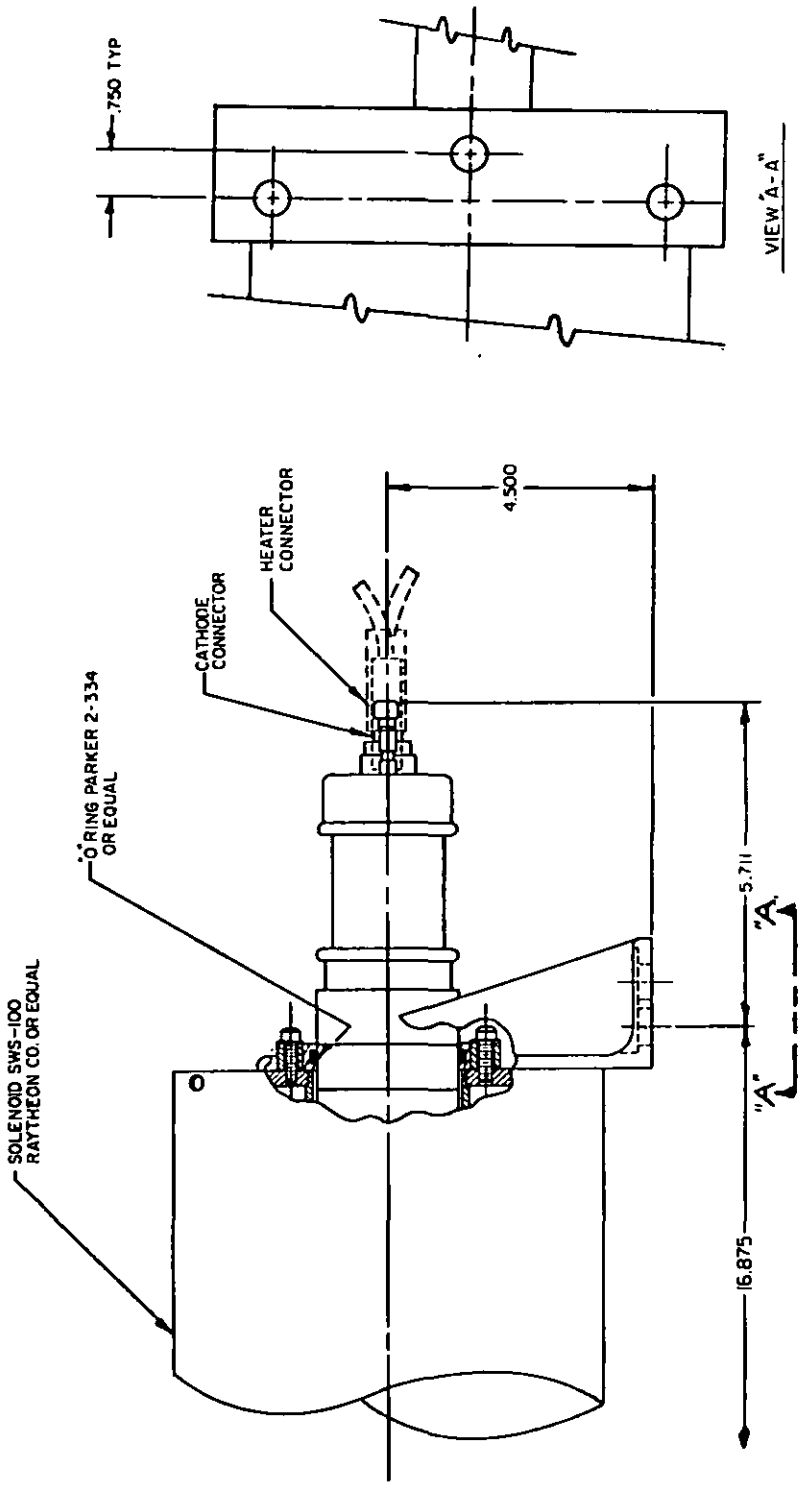
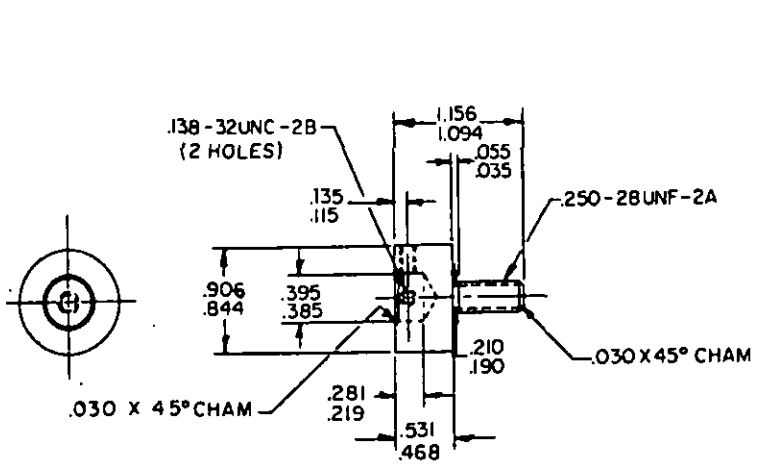
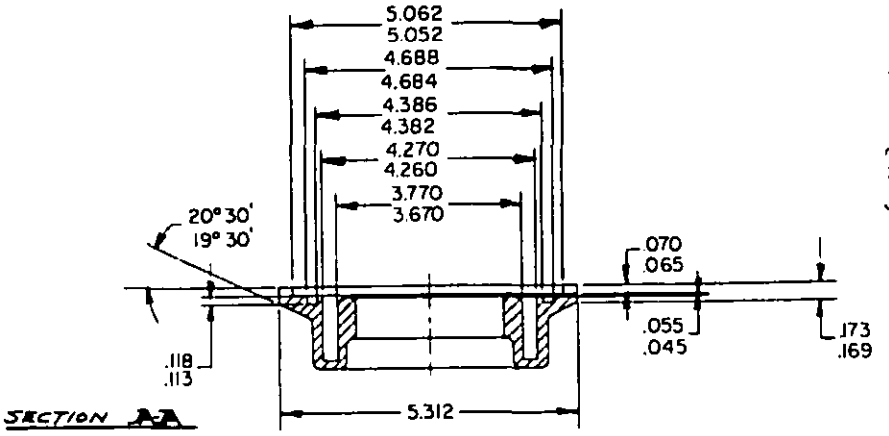


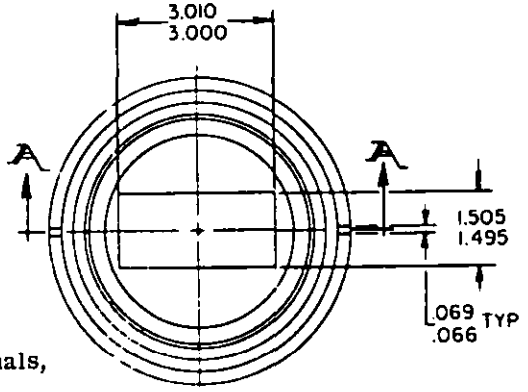
FIGURE 3. Installation drawing (for guidance only) - Continued.

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DETAIL OF ADAPTER (GROUND)



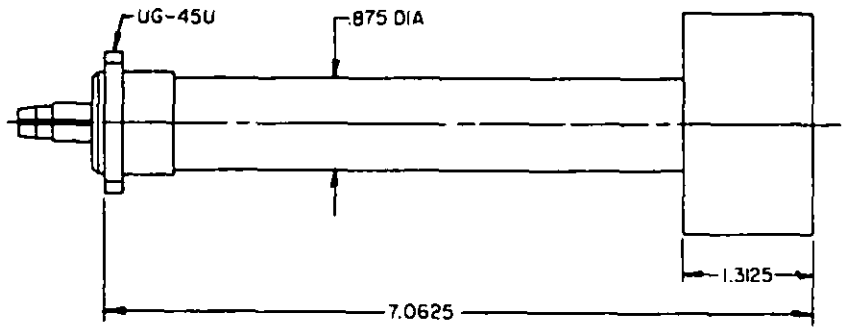


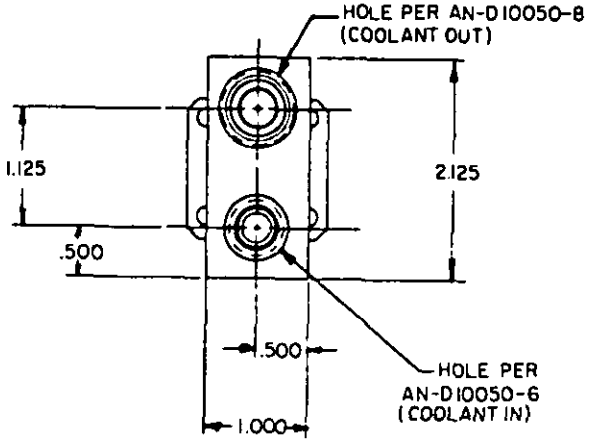
WAVEGUIDE CHOKE FLANGE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) for three place decimals, tolerance is ±.02 (.51 mm) for two place decimals.

FIGURE 3. Installation drawing (for guidance only) - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.030	.76	.210	5.33	1.094	27.79	4.386	111.40
.035	.89	.219	5.56	1.125	28.58	4.500	114.30
.045	1.14	.250	6.35	1.156	29.36	4.684	118.97
.055	1.40	.281	7.14	1.3125	33.34	4.688	119.08
.065	1.65	.385	9.78	1.495	37.97	5.052	128.32
.066	1.68	.395	10.03	1.505	38.23	5.062	128.57
.069	1.75	.469	11.91	2.125	53.98	5.312	134.92
.070	1.78	.500	12.70	3.000	76.20	5.711	145.06
.113	2.87	.531	13.49	3.010	76.45	7.0625	179.39
.115	2.92	.625	15.88	3.375	85.73	8.750	222.25
.118	3.00	.750	19.05	3.760	95.50	9.375	238.13
.135	3.43	.844	21.44	3.770	95.76	12.685	322.20
.169	4.29	.875	22.23	4.260	108.20	16.875	428.63
.173	4.39	.906	23.01	4.270	108.46		
.190	4.83	1.000	25.40	4.382	111.30		

TERMINATION Lc SH 1
Flow = Water, 1 G.P.M.
Pressure drop = 10 P.S.I.
Pressure max = 115 P.S.I.A.

FIGURE 3. Installation drawing (for guidance only) - Continued.

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